UC San Diego
RADY SCHOOL OF MANAGEMENT

MGT 451R

AI & Disruptive Technology Strategy

(Previous Title: Technology & Innovation Strategy)

Syllabus

(significantly updated)

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COURSE DESCRIPTION

How will AI redefine businesses and enable industry transformation in the digital age?

To address this question, we will study the relevant frameworks and tools for formulating AI (Artificial Intelligence) innovation, disruptive technologies, and digital transformation. It aims to help students gain insights into technology change and business strategy by providing a future-focused lens to how competitive advantage is achieved through innovation, so as to better prepare students to become leaders in the tech-intensive world.

Specifically, this course concentrates on understanding the fundamental topics of innovation in both established firms and startups as driven by AI and related technologies. More specifically, we will:

- ✓ begin with the S curve, a simple but powerful tool to chart disruptive technologies for leadership, startups, investment, and careers;
- ✓ exam how disruptive technologies (AI in particular) create opportunities for new businesses and transform established ones;
- ✓ introduce the Technology-Organization-Ecosystem (TOE) theory, distilled from original research;
- ✓ learn how to make decisions about investment in technologies and new ventures;
- ✓ zoom on emerging technologies (e.g., Generative AI, LLM, embodied AI, robotics, FSD, data analytics, Fin Tech, block chain, digital currency…);
- ✓ go through key topics such as AI+X, digital transformation, platform leadership, tech standards, user-centric innovation, technology finance, real options, social media & marketing, and digital healthcare.
- ✓ look into the movers, shakers, and game changers along the new S curves in many industries redefined by AI and disruptive technologies.

FORMAT

This non-traditional course is mostly online and **asynchronous**, so you can watch the lecture videos and participate in online discussions at your own pace, with weekly due dates and milestones along the way. Some classes may be enhanced with hybrid sessions. Overall, it offers the optimal combination of flexibility and interactive learning in the digital age. It's a new S curve in learning.

OBJECTIVES

- ✓ Learn to think like a technology leader with vision and foresight;
- ✓ Get inside disruptive technologies (AI in particular) and how they will offer the best opportunities (along with the challenges) for business leaders;
- ✓ Learn the TOE theory and S curves to discriminate between trends and fads;
- ✓ Learn to make better decisions on technologies, products, markets, and investments;
- ✓ Identify and evaluate the transformative impact of AI technologies across industries such as hi-tech, finance, healthcare, and education, focusing on their potential to drive innovation and productivity;
- ✓ Assess the social, ethical and legal implications of AI, developing the skills to navigate challenges related to governance, security and accountability;
- ✓ Propose innovative AI initiatives or startups that address industry pain points or unmet needs, demonstrating their strategic value and feasibility;
- ✓ Get positioned into great career paths of a venture capitalist, technology leader, entrepreneur, consultant, investment banker, technology analyst, product manager, financial analyst, or whatever career you'd like to pursue as long as innovation is part of the skill set.

AUDIENCE

- ✓ those who anticipate taking management positions (e.g., CEOs, CTOs, CIOs, VP for strategy, product manager, marketing manager, etc) in both start-ups and established firms where they must formulate strategy;
- ✓ those who anticipate investing in technology markets and must analyze firm strategy;
- ✓ those who anticipate contracting or consulting with firms that do much of their business in technology markets;
- ✓ those who work in accounting, finance, operations, and supply chains;
- ✓ anyone who cares about the future.

Companies to be discussed:

- ✓ **Industry leaders**: Tesla, Nvidia, Qualcomm, Goldman Sachs, Netflix, Amazon…
- ✓ **Game changers:** Open AI, Waymo, ARK Invest, Palantir, Intuitive Surgical, DeepMind, Dexcom…
- ✓ **Many movers & shakers**: Joby Aviation, Butterfly, Zebra, AiDoc, Bit Coins…

COURSE MATERIALS

- ✓ Most readings and cases are put together by the Professor, and are accessible online through Canvas. Some materials may require log-in from university's VPN.
- ✓ No required textbooks.

ASSIGNMENTS & GRADING

- ✓ Weekly online quizzes about video lectures and readings;
- ✓ Weekly case study and discussions;

- ✓ One case presentation (1 per group throughout the quarter, 10~15 minutes each);
- ✓ One term project presentation (1 per group, in lieu of final exam).

Grading Policy:

Category	Assignment	Group/Individual	Weight
Lectures/Readings	Video lectures/readings Weekly online quizzes	Individual	30%
Participation	Active participation in discussions (online+offline)	Individual	30%
1 Case Presentation	Submit slides and recorded videos (1 per group)	Group	20%
1 Term Project	Topic selection, proposal, presentation with slides (written report not required)	Group	20%
Exams?	No midterm or final exams		0

Note:

- **Updated** this year: This is the latest and accurate. If you see any mentioning elsewhere different from this table, please trust this one. The course workload has been adjusted toward quality over quantity. For example, we no longer require project written reports, case writeups, midterm or final exams (which were required in previous years). Hope this helps you to be more focused on the most important learnings.
- **Group**: You need to form a group early. Each group should consist of X members (X is the specific size to be announced ahead of each class, pending on enrollments). For different group size you'll need to get in advance approval from the Professor or TA.

AI & Disruptive Tech Strategy

Weekly Schedule, Topics, Cases and Readings

Session 1: Overview

Topics:

- The Syllabus
- Course framework: Technology and innovation in the AI era
- Technology, strategy, market, and competition: Strategic and economic perspectives
- Impacts on industries: AI as transformational agent

Lecture:

- Join the real-time session on Zoom
- Watch the lecture videos, and answer the quiz questions online

Case Study: ***AI for Business***

- "Realizing the Generative AI Opportunity: Embracing Change to Create Business Value," Harvard Business Review

Readings:

- The economic potential of generative AI: The next productivity frontier
- Read the Syllabus (and answer the quizzes about the Syllabus and Academic Integrity)
- Preview next session's readings

Session 2: AI & Technology Frontiers

Topics:

- AI frontiers
- Agentic AI
- Generative AI, LLM, ChatGPT
- Sensors, computer vision, image recognition…
- Technology-enabled startups in the digital-age ...
- Solopreneurs

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: ***Technology Frontiers: Agentic AI***

- Why agents are the next frontier of generative AI, McKinsey & Company

Reading:

- What is an AI agent? McKinsey Quarterly

- Seizing the Agentic AI Advantage, AI by McKinsey

Session 3: S-Curves and TOE Theory for Technology Strategy

Topics:
- Deep dive into foundational theory
- Featured frameworks to understand technology change
- Economics of technological evolution, Technology lifecycle
- S-Curves
- TOE Theory
- The TOE analysis of AI industry leaders

Lecture:
- Watch the lecture videos, and answer the quiz questions online

Case Study: ***TOE Theory***
- The Process of Innovation Assimilation by Firms in Different Countries: A Technology Diffusion Perspective, by Kevin Zhu et al, Management Science [PDF]

Readings:
- A Theory of Rapid Transition: How S-Curves Work

Session 4: Digital Transformation by Incumbent Companies

Topics:
- Technological change induces *transformation* (or *failure*) of incumbent firms
- Creative destruction
- Transformation of incumbents or upgrade current businesses: TOE upgrade
- Focus: How is Gen AI transforming financial services

Lecture:
- Watch the lecture videos, and answer the quiz questions online

Case Study: ***AI-enabled Transformation in Financial Services***
- Goldman Sachs: Scaling Generative AI in Financial Services [PDF]

Readings:
- One year in: Lessons learned in scaling up generative AI for financial services
- "Innovation diffusion in global contexts: determinants of post-adoption digital transformation of European companies." By Kevin Zhu et al, *European Journal of Information Systems*. [PDF]

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Session 5: Product vs Platform Leadership by Innovators

Topics:

- Disruptive technologies: Opportunities for new businesses
- Commercialization of new technologies
- How to capture value from technological innovations (often in the face of competition)?
- IP strategy: open or closed?
- Platform-based competition

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: *Platform Leader*

- NVIDIA and the CUDA Platform: Platform Leader Powering the AI Ecosystem [PDF]

Readings:

- The elements of platform leadership, *MIT Sloan Mgmt Review* (accessible with university's VPN)
- Intellectual property strategy and the governance of technological platform-driven global value chains: The case of Qualcomm

Session 6: Standards-based Ecosystems

Topics:

- The role of standards in technology development and diffusion
- Network effects, lock-in, and switching costs in standards-oriented markets
- Standards war, platform battles
- How will technology standards define the future of mobility as service…

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: *Technology Standards*

- Tesla vs. Waymo: A Standards War in Robotaxis

Readings:

- "The Art of Standards Wars," by Carl Shapiro and Hal Varian, *California Management Review*.
- Electric Vehicle Charging: The Role of SAE and Tesla's North American Charging Standard

Session 7: Generative AI to Transform Healthcare Services

Topics:

- How digital technologies finally ready to revolutionize the way we manage healthcare
- LLM in healthcare
- DNA, genomics, personalized medicine
- New trends: wireless health, mobile health, preventive care, home care…
- Biotech, telemedicine, bio-informatics, digital medical records…

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: ***Gen AI to Transform Healthcare***

- Mayo Clinic Platform: Harnessing Disruptive AI for System-Wide Transformation

Readings:

- Generative AI in healthcare: an implementation science informed translational path on application, integration and governance

Session 8: Disruptive Technologies to Transform Medicine

Topics:

- Life sciences meet information sciences
- Robotics in healthcare
- Telemedicine, telesurgery….

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: ***Intuitive Surgical***

- Intuitive Surgical and Its da Vinci Platform: AI, TOE, and the Future of Robotic Surgery

Readings:

- Scaling gen AI in the life sciences industry, McKinsey Report
- The robotic surgery market battle is heating up, MedTech Dive

Session 9: Investment in Disruptive Technologies

Topics:

- How to invest in disruptive technologies or startups with game-changing potential, but unproven cash flows, high risks and great uncertainty?
- Strategic investment in disruptive technologies (e.g., AI)
- Decision making about technology projects
- Capex and ROI

- Real options
- Data analytics for decision-making

Lecture:

- Watch the lecture videos, and answer the quiz questions online

Case Study: *ARK Invest*

- ARK Invest: Betting Big on Disruptive Technologies That Will Define the Future [PDF]

Readings:

- Evaluating Technology Investment Projects: The Real Options Approach, by Kevin Zhu [PDF]
- How Archer Aviation's 214% Stock Surge Stacks Up After Stellantis Partnership News
- Read "Flying taxis are on the horizon as aviation soars into a new frontier"
- Question for you: Would you invest in (or join) such a startup (e.g., Joby Aviation) or other disruptive startups we covered throughout this course? What's your most favorite?

Session 10: Real-World Applications & Projects